THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 4: **Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Most alpha particles pass undeflected through a thin gold foil. Reason (R): The nucleus occupies a very small fraction of the atom's volume. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. An ion has 18 electrons and a nucleus with 16 protons. The ion is: 1

(a) S^{2-} **(b)** Cl^{-} **(c)** S^{2+} **(d)** Ar

3. An unknown element has isotopes 24 (79 per cent), 25 (10 per cent) and 26 (11 per cent). The average atomic mass is: 1

(a) 24 u **(b)** 24.9 u **(c)** 24.3 u **(d)** 25 u

4. Assertion (A): Isotopes of an element have similar chemical properties. Reason (R): Isotopes have the same number of electrons. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. Two ions X^{2+} and Y^{2-} each have 18 electrons. The atomic numbers of X and Y are respectively: 1

(a) 16 and 20 **(b)** 20 and 16 **(c)** 19 and 17 **(d)** 18 and 18

6. A sample of boron (average mass 10.8 u) has isotopes 10 and 11. The percentage of the lighter isotope is: 1

(a) 20 **(b)** 50 **(c)** 80 **(d)** 10.8

7. Assertion (A): Chlorine has an atomic mass of 35.5 u. Reason (R): Chlorine is made of two isotopes with whole-number masses 35 u and 37 u in a 3 : 1 ratio. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. The atomic number of an element is 17 and its ion has a charge of -1. The ion's configuration is the same as that of: 1

(a) helium **(b)** krypton **(c)** neon **(d)** argon

9. **Assertion (A):** In Rutherford's experiment most alpha particles passed straight through the gold foil. 1
Reason (R): Most of the space inside an atom is empty.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Rutherford's nuclear model could not explain the stability of the atom. 1
Reason (R): The neutron was discovered by James Chadwick in 1932.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the percentage of the heavier isotope for an element of average mass 63.5 u with isotopes 63 u and 65 u. 2

12. State the reason why the first shell is filled with only two electrons. 2

13. State two reasons why Thomson's model could not explain Rutherford's results. 2

14. Why is the atomic mass of most elements not a whole number although their mass numbers are? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A metal M forms an oxide M_2O_3. Its atom has 13 protons. State its configuration, ion and the formula unit mass if its mass number is 27. 3

16. An atom has 5 more neutrons than protons and its mass number is 31. Find Z, n and the electronic configuration. 3

17. Explain how the Bohr model accounts for the observation that electrons do not fall into the nucleus. 3

OR

Explain, using Bohr's idea, why the elements in the same group show similar chemistry and the elements in the same period show a gradual change. 3

18. Explain the use of uranium-235 in nuclear reactors and the safety concerns. 3

OR

An atom loses two electrons and the resulting ion has configuration 2, 8, 8. Find the atomic number and the name. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** An atom of an element X has 11 protons and 12 neutrons. Meena is asked to describe it using what she has learnt about atomic structure. 5
(i) Find the atomic number and the mass number of X.
(ii) Write its electronic configuration.
(iii) What is its valency?
(iv) What happens to the electronic configuration when the atom forms an ion of valency 1, and how many electrons does the ion have?

20. Explain, with reference to the gold foil experiment, how the nuclear radius can be estimated and why the nucleus is so dense (nuclear radius about 10^-15 m, mass about 10^-26 kg). 5

OR

Discuss the idea that 'the structure of the atom explains the periodic similarity in chemical behaviour'. Use Li, Na, K and F, Cl, Br as examples. 5

END OF QUESTION PAPER — Total: 40 Marks